ALGORITHM: CSV TO JSON CONVERTER

Step 1: Start.

Step 2: Import the **csv** and **json** modules.

Step 3: Define the **read_csv(input_path)** function.

Step 4: Open the CSV file in read mode.

Step 5: Use **csv.DictReader()** to read the CSV file and use the first row as dictionary keys.

Step 6: Convert all CSV rows into a list of dictionaries and return the list.

Step 7: Define the **write_json(output_path, data)** function.

Step 8: Open/create the JSON file in write mode.

Step 9: Use **json.dump()** to save the list of dictionaries as formatted JSON.

Step 10: Define the **convert_csv_to_json(input_path, output_path)** function.

Step 11: Read the CSV data using **read_csv()**.

Step 12: Write the data to the JSON file using **write_json()**.

Step 13: In the main program, set the input file as **students.csv** and output file as **students.json**.

Step 14: Create a sample CSV file containing student records.

Step 15: Display the contents of the CSV file.

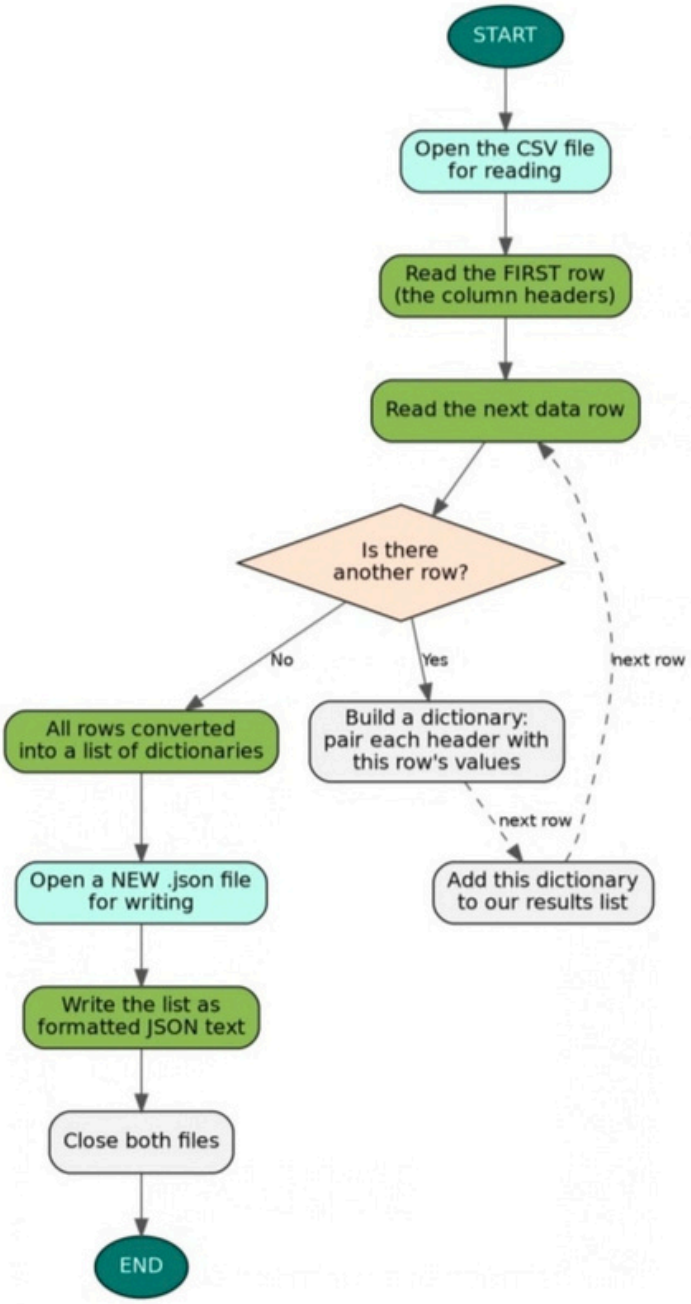
Step 16: Convert the CSV data into JSON.

Step 17: Display the number of converted rows and the JSON file location.

Step 18: Display the contents of the resulting JSON file.

Step 19: Stop.

FLOWCHART


START
Open the CSV file for reading
Read the FIRST row (the column headers)
Read the next data row
Is there another row?
No
Yes
next row
All rows converted into a list of dictionaries
Build a dictionary: pair each header with this row's values
next row
Open a NEW .json file for writing
Add this dictionary to our results list
Write the list as formatted JSON text
Close both files
END

Code

CSV TO JSON CONVERTER — FULL CODE

```
"""
============================================================
CSV TO JSON CONVERTER

This script reads a CSV file, converts every row into a JSON
record using the header row as the labels, and saves the result
as a brand-new .json file.
============================================================
"""

import csv
import json


# ============================================================
# 1. READ THE CSV FILE
# ============================================================
def read_csv(input_path):
    """Returns a list of dictionaries, one dictionary per CSV row."""
    with open(input_path, "r", newline="") as f:
        reader = csv.DictReader(f)
        return list(reader)


# ============================================================
# 2. WRITE THE DATA AS JSON
# ============================================================
def write_json(output_path, data):
    """Writes 'data' to output_path as formatted JSON."""
    with open(output_path, "w") as f:
        json.dump(data, f, indent=4)


# ============================================================
# 3. MAIN CONVERSION FUNCTION
# ============================================================
def convert_csv_to_json(input_path, output_path):
    """Reads a CSV file and writes its contents as JSON."""
    data = read_csv(input_path)
    write_json(output_path, data)
    return data


# ============================================================
# 4. DEMO / DRIVER CODE
# ============================================================
if __name__ == "__main__":

    input_path = "students.csv"
    output_path = "students.json"

    # --- create a sample CSV file, just for this demo ---
    sample_csv = (
        "id,name,department,marks\n"
```

```
    "1,Aditi Sharma,Computer Science,88\n"
    "2,Rahul Verma,Mechanical,76\n"
    "3,Sneha Iyer,Electronics,92\n"
)

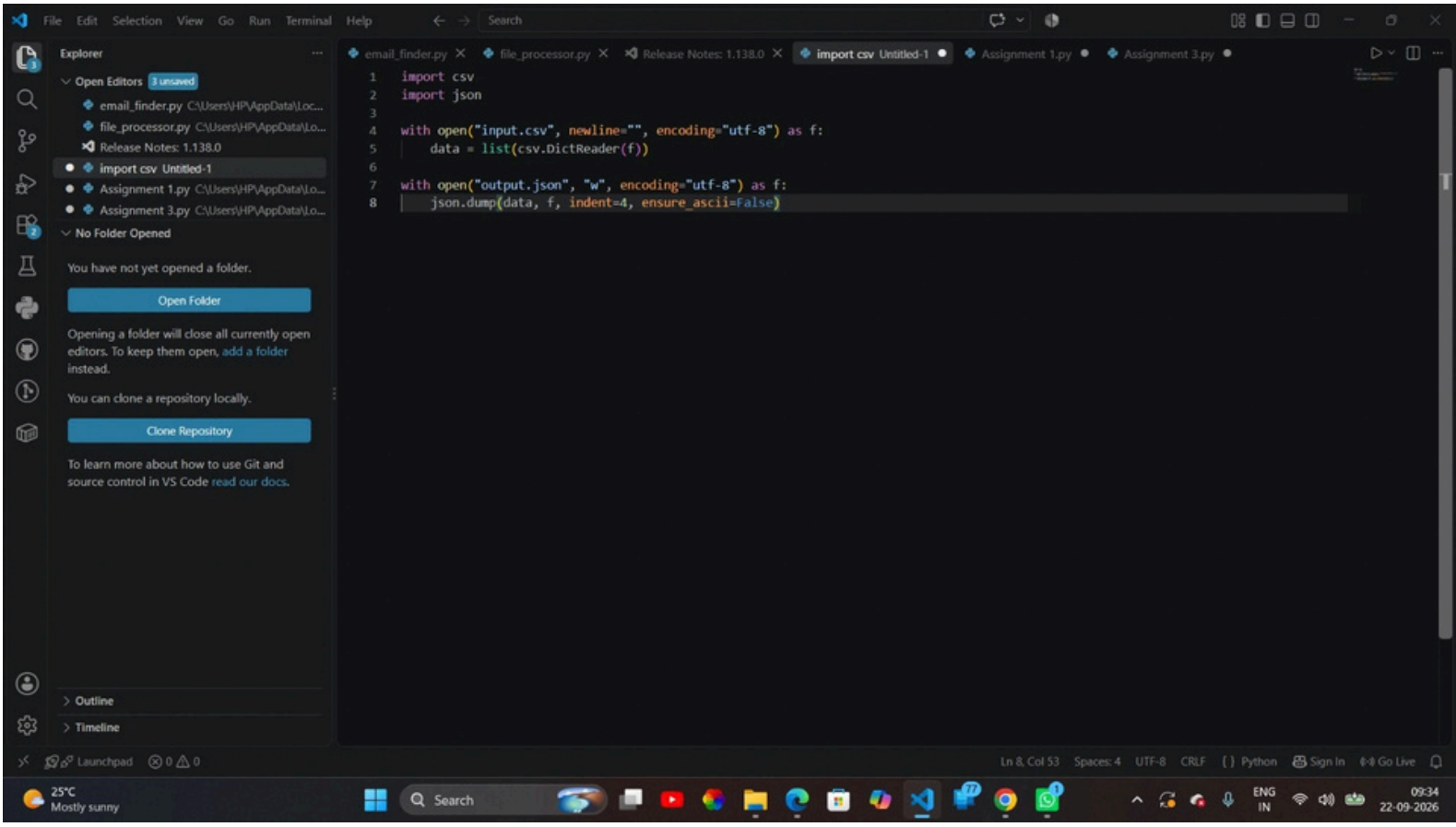
with open(input_path, "w", newline="") as f:
    f.write(sample_csv)

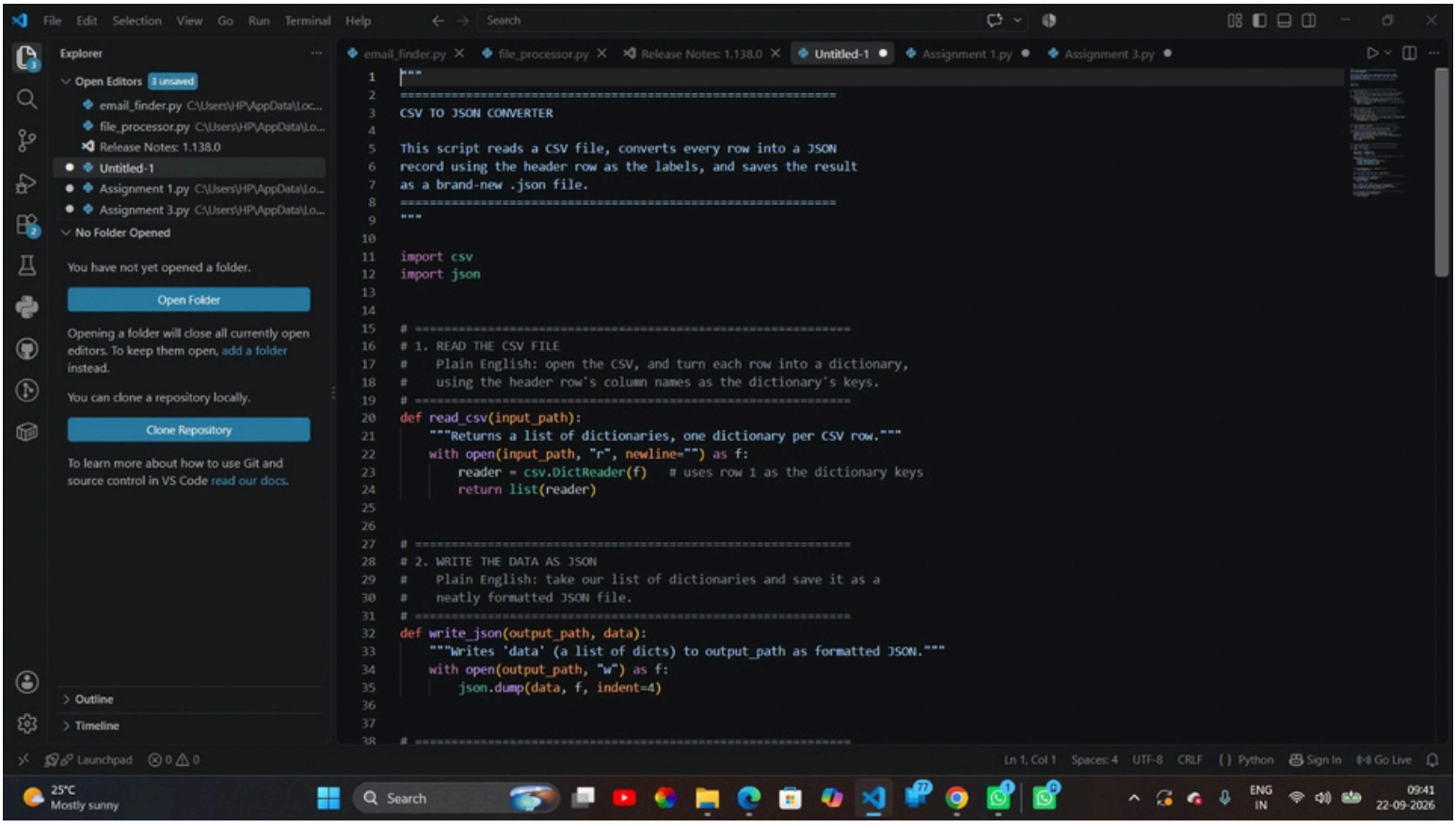
print(f"Created sample CSV file: {input_path}\n")

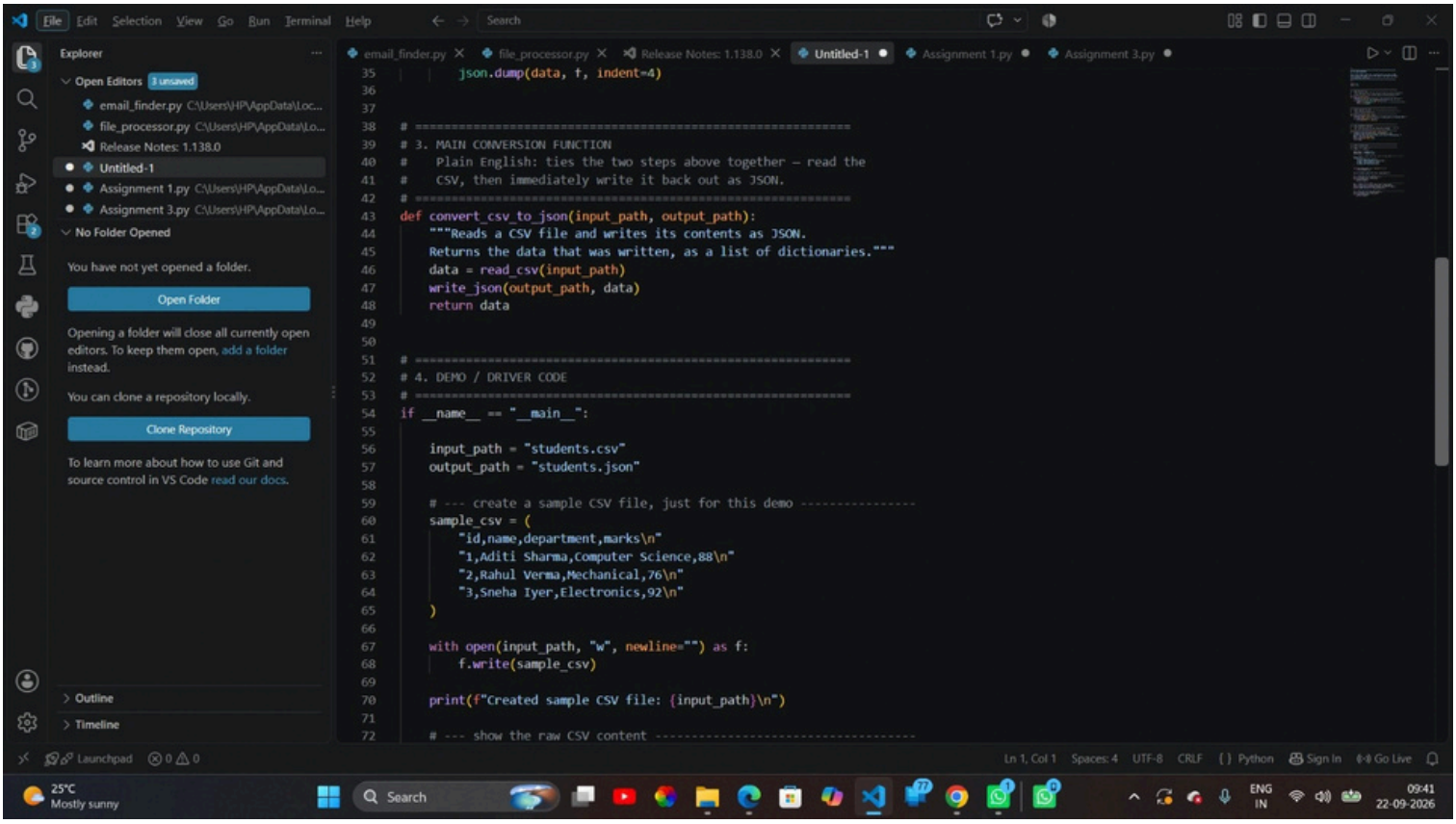
# --- show the raw CSV content ---
print(f"Contents of '{input_path}':")
with open(input_path, "r") as f:
    print(f.read())

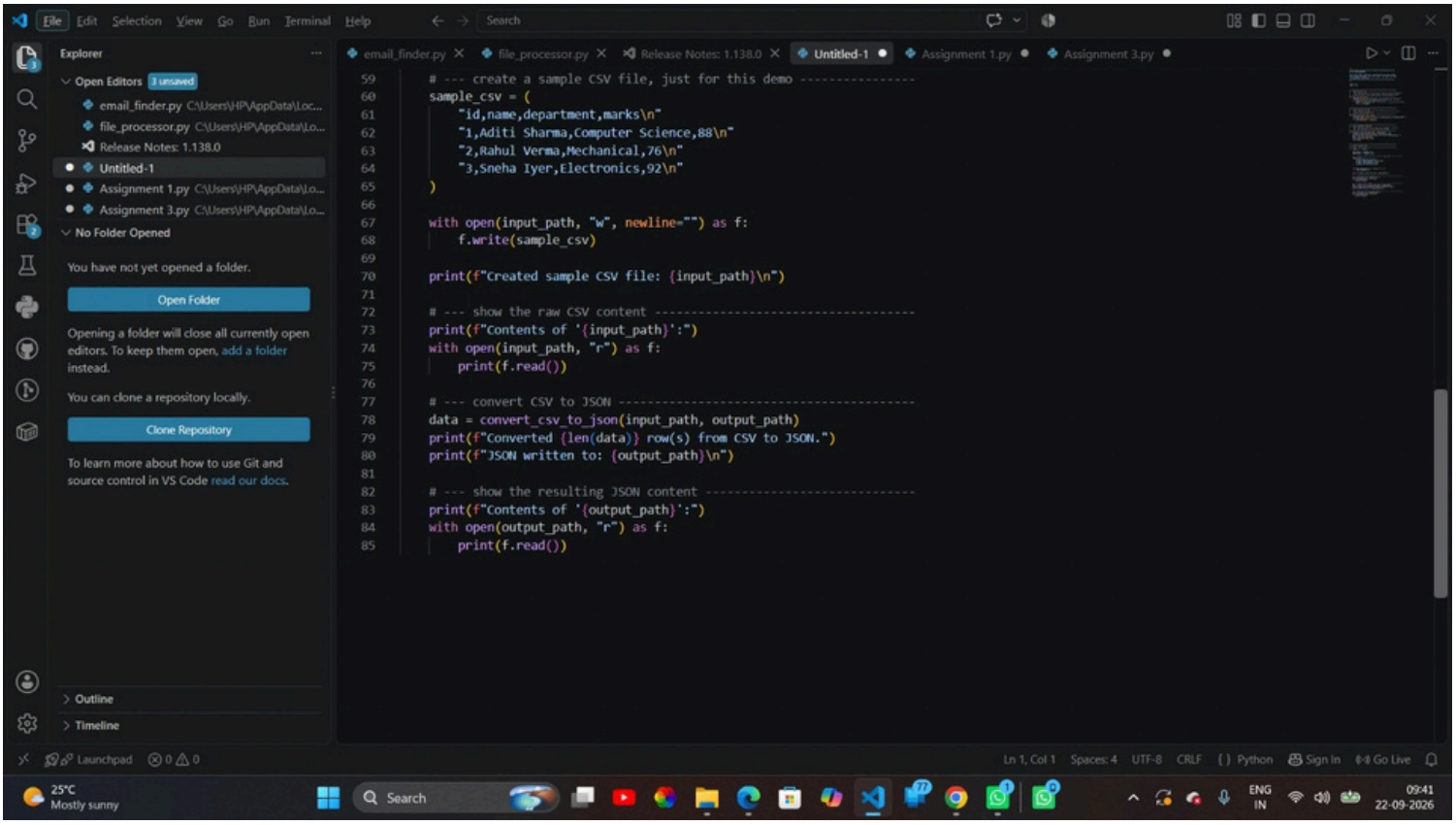
# --- convert CSV to JSON ---
data = convert_csv_to_json(input_path, output_path)
print(f"Converted {len(data)} row(s) from CSV to JSON.")
print(f"JSON written to: {output_path}\n")

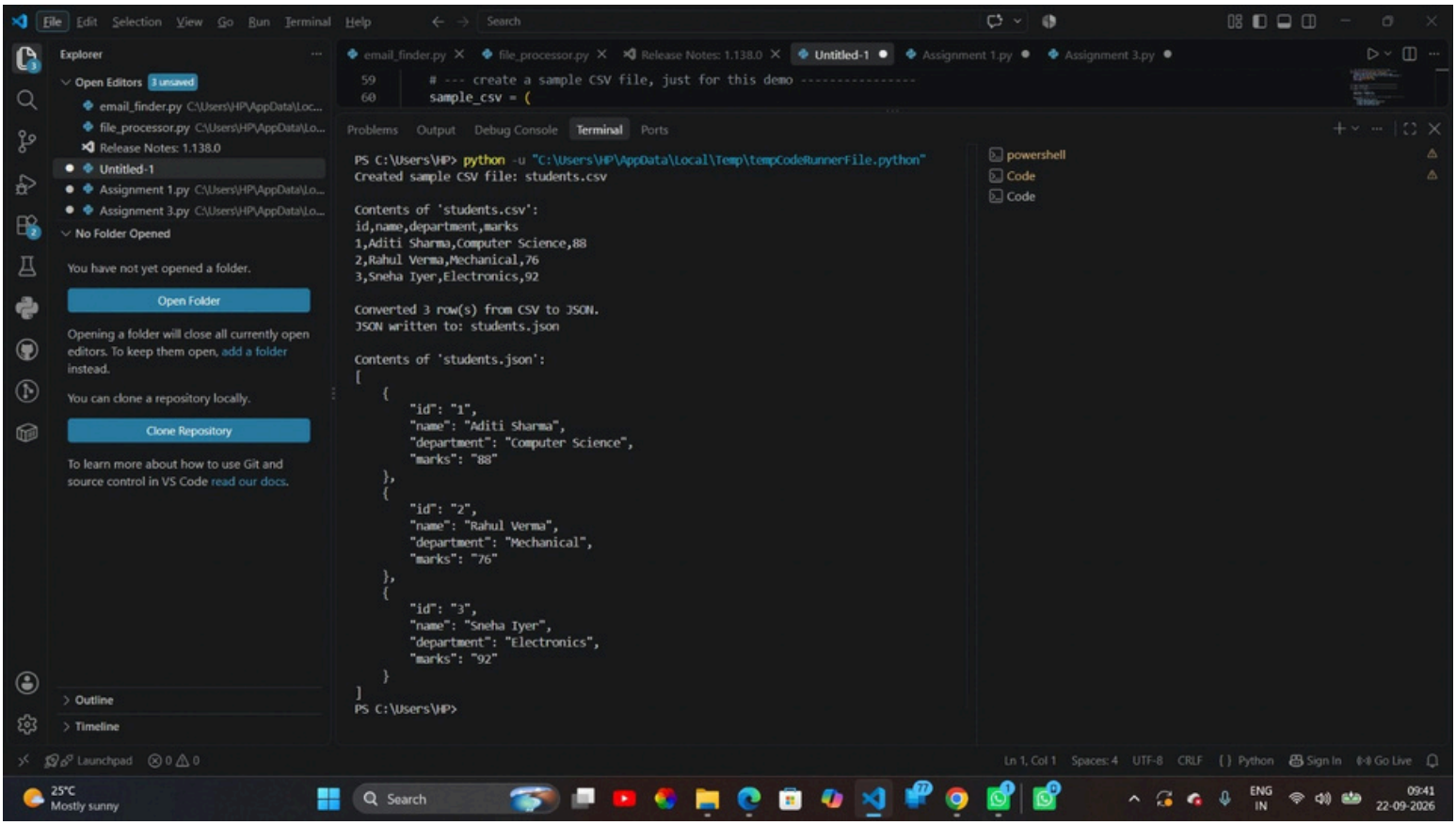
# --- show the resulting JSON content ---
print(f"Contents of '{output_path}':")
with open(output_path, "r") as f:
    print(f.read())
```


import csv
import json
with open("input.csv", newline="", encoding="utf-8") as f:
data = list(csv.DictReader(f))
with open("output.json", "w", encoding="utf-8") as f:
json.dump(data, f, indent=4, ensure_ascii=False)


"""
CSV TO JSON CONVERTER
This script reads a CSV file, converts every row into a JSON
record using the header row as the labels, and saves the result
as a brand-new .json file.
"""
import csv
import json
1. READ THE CSV FILE
Plain English: open the CSV, and turn each row into a dictionary,
using the header row's column names as the dictionary's keys.
def read_csv(input_path):
"""Returns a list of dictionaries, one dictionary per CSV row."""
with open(input_path, "r", newline="") as f:
reader = csv.DictReader(f) # uses row 1 as the dictionary keys
return list(reader)
2. WRITE THE DATA AS JSON
Plain English: take our list of dictionaries and save it as a
neatly formatted JSON file.
def write_json(output_path, data):
"""Writes 'data' (a list of dicts) to output_path as formatted JSON."""
with open(output_path, "w") as f:
json.dump(data, f, indent=4)


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email_finder.py C:\Users\HP\AppData\Loc...
file_processor.py C:\Users\HP\AppData\Lo...
Release Notes: 1.138.0
Untitled-1
Assignment 1.py C:\Users\HP\AppData\Lo...
Assignment 3.py C:\Users\HP\AppData\Lo...
No Folder Opened
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
You can clone a repository locally.
Clone Repository
To learn more about how to use Git and source control in VS Code read our docs.
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email_finder.py
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Release Notes: 1.138.0
Untitled-1
Assignment 1.py
Assignment 3.py
59 # --- create a sample CSV file, just for this demo ----------
60 sample_csv = (
Problems Output Debug Console Terminal Ports
PS C:\Users\HP> python -u "C:\Users\HP\AppData\Local\Temp\tempCodeRunnerFile.python"
Created sample CSV file: students.csv
Contents of 'students.csv':
id,name,department,marks
1,Aditi Sharma,Computer Science,88
2,Rahul Verma,Mechanical,76
3,Sneha Iyer,Electronics,92
Converted 3 row(s) from CSV to JSON.
JSON written to: students.json
Contents of 'students.json':
[
{
"id": "1",
"name": "Aditi Sharma",
"department": "Computer Science",
"marks": "88"
},
{
"id": "2",
"name": "Rahul Verma",
"department": "Mechanical",
"marks": "76"
},
{
"id": "3",
"name": "Sneha Iyer",
"department": "Electronics",
"marks": "92"
}
]
PS C:\Users\HP>
powershell
Code
Code
Launchpad 0 0
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Python Sign In Go Live
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